

Package ‘RLFPCR’

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Type Package

Title Robust Longitudinal Functional Principal Component Regression

Version 1.0.0

Description Functions for fitting robust longitudinal functional principal component regression models for longitudinal scalar-on-function data. The package provides tools for simulation-based data generation, robust estimation of longitudinal functional principal component decompositions, robust mean-surface and covariance estimation, separation of subject-level functional intercept, subject-level functional slope, and visit-specific deviation components, blockwise estimation of longitudinal functional principal component scores, robust additive mixed score regression, and coefficient-function reconstruction. The implemented methodology is designed for repeatedly observed scalar responses and densely observed functional predictors, with resistance to atypical curves, contaminated visits, and outlying scalar responses.

Depends R ($\geq 3.5.0$)

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stats

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Encoding UTF-8

LazyData true

ByteCompile true

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fit_rlfpcr

*Fit Robust Longitudinal Functional Principal Component Regression***Description**

Fits the second-stage robust longitudinal functional principal component regression model. The function uses robust longitudinal FPCA scores obtained from `rlfpca_estimate` and fits a robust additive mixed score regression for a repeated scalar response.

Usage

```
fit_rlfpcr(y, lfpca, subject, times,
           scalar_cov = NULL,
           max_iter = 4,
           psi = "bisquare",
           c = 4.685)
```

Arguments

y	A numeric vector containing the repeated scalar response values. Its length must be equal to the total number of subject-visit observations.
lfpca	An object returned by <code>rlfpca_estimate</code> . It must contain the robust longitudinal FPCA scores and eigenfunctions, including at least <code>xi</code> , <code>zeta</code> , <code>phi.0</code> , <code>phi.1</code> , and <code>phi.U</code> .
subject	A vector of subject identifiers. The length of <code>subject</code> must be the same as the length of <code>y</code> . Repeated observations belonging to the same subject must share the same identifier.
times	A numeric vector of visit times. The length of <code>times</code> must be the same as the length of <code>y</code> . The time variable is used both in the smooth population time effect and in the interaction between subject-level scores and visit time.
scalar_cov	An optional matrix or data frame of scalar explanatory variables. The number of rows must be equal to the length of <code>y</code> . If column names are not supplied, default names <code>q1</code> , <code>q2</code> , ... are assigned internally. The default is <code>NULL</code> .
max_iter	Maximum number of iteratively reweighted least-squares iterations used in the robust score-space regression. The default is 4.
psi	Type of robust score/weight function used for the scalar-response residuals. Currently supported choices are "bisquare" and "huber", as implemented by the internal weight function. The default is "bisquare".
c	Tuning constant for the robust weight function. The default value is 4.685, corresponding to the usual Tukey bisquare tuning constant.

Details

The function implements the final regression stage of robust longitudinal functional principal component regression (RLFPCR). Let Y_{ij} denote the scalar response for subject i at visit j , let T_{ij} denote the visit time, and let $X_{ij}(s)$ denote the longitudinal functional predictor. The underlying scalar-on-function model is

$$Y_{ij} = \alpha(T_{ij}) + q_{ij}^\top \delta + r_{ij}^\top u_i + \int_{\mathcal{S}} \{X_{ij}(s) - \mu(s, T_{ij})\} \beta(s) ds + e_{ij}.$$

The robust longitudinal FPCA step decomposes the centered functional predictor as

$$X_{ij}(s) - \mu(s, T_{ij}) \approx \sum_{k=1}^{K_X} \xi_{ik} \{ \phi_k^{(0)}(s) + t_{ij} \phi_k^{(1)}(s) \} + \sum_{\ell=1}^{K_U} \zeta_{ij\ell} \phi_\ell^U(s),$$

where ξ_{ik} are subject-level scores, $\zeta_{ij\ell}$ are visit-level scores, $\phi_k^{(0)}$ and $\phi_k^{(1)}$ describe the subject-level baseline and temporal components, and ϕ_ℓ^U describes the visit-level deviation component.

Substitution of this truncated representation into the scalar-on-function model gives the finite-dimensional score regression

$$Y_{ij} = \alpha(T_{ij}) + q_{ij}^\top \delta + r_{ij}^\top u_i + \sum_{k=1}^{K_X} a_k \xi_{ik} + \sum_{k=1}^{K_X} b_k t_{ij} \xi_{ik} + \sum_{\ell=1}^{K_U} c_\ell \zeta_{ij\ell} + e_{ij}.$$

The function fits this model using `gam`. The smooth population time effect is represented by `s(times, bs = "ps", m = 1)`, and subject-specific dependence in the scalar response is represented by the random-effect smooth `s(subject, bs = "re")`. Robustness is introduced by iteratively updating observation weights according to

$$w_{ij} = \frac{\psi(r_{ij}/\hat{\tau})}{r_{ij}/\hat{\tau}},$$

where r_{ij} is the scalar-response residual and $\hat{\tau}$ is a MAD-type robust scale estimate.

After estimating the score coefficients, the functional coefficient is reconstructed as

$$\hat{\beta}(s) = \sum_{k=1}^{K_X} \hat{a}_k \hat{\phi}_k^{(0)}(s) + \sum_{k=1}^{K_X} \hat{b}_k \hat{\phi}_k^{(1)}(s) + \sum_{\ell=1}^{K_U} \hat{c}_\ell \hat{\phi}_\ell^U(s).$$

Thus, the returned coefficient function decomposes the functional effect into a subject-level baseline component, a subject-level longitudinal/time component, and a visit-level deviation component.

Value

A list with the following components:

<code>fit</code>	The final weighted <code>gam</code> object fitted after the robust IRLS iterations.
<code>beta_hat</code>	A numeric vector containing the reconstructed functional coefficient $\hat{\beta}(s)$ evaluated on the same functional grid as the eigenfunctions stored in <code>lfpca</code> .
<code>delta</code>	A numeric vector of estimated coefficients for the scalar explanatory variables. If <code>scalar_cov = NULL</code> , this is a numeric vector of length zero.
<code>a</code>	Estimated coefficients corresponding to the subject-level baseline scores ξ_{ik} .
<code>b</code>	Estimated coefficients corresponding to the subject-level longitudinal score interactions $t_{ij}\xi_{ik}$.
<code>cU</code>	Estimated coefficients corresponding to the visit-level deviation scores $\zeta_{ij\ell}$.
<code>weights</code>	Final robust observation weights from the IRLS procedure.
<code>design</code>	The data frame used internally to fit the score-space regression model. It contains the scalar response, visit times, subject indicator, optional scalar explanatory variables, and the constructed longitudinal FPCA score covariates.

Note

This function should usually be applied after `rlfpca_estimate`. The theoretical development assumes fixed truncation dimensions, whereas the default implementation of `rlfpca_estimate` may select the number of components by an explained-variance criterion.

Examples

```

dat <- simulate_data(
  case = "1",
  nI = 100,
  nV = 4,
  tps = seq(0.5, 99.5, by = 1),
  sigma_x = 0.01,
  sigma_y = sqrt(2),
  sigma_b = sqrt(2),
  scalar_cov = 4,
  beta_scalar = c(0.6, 0.8, -0.5, 0.2),
  outlier = TRUE
)

X <- dat$X
y <- dat$y
subject <- dat$subject
times <- dat$times
tps <- dat$tps
scalar_cov <- dat$scalar_cov

lfpca <- rlfpcr_estimate(
  Y = X,
  subject = subject,
  Time = times,
  L = 0.95,
  smooth = TRUE,
  bf = 8,
  df_time_mean = 5,
  mean_iter = 3,
  clean_cells = TRUE,
  clean_c = 4.5,
  score_iter = 8
)

fit <- fit_rlfpcr(
  y = y,
  lfpca = lfpca,
  subject = subject,
  times = times,
  scalar_cov = scalar_cov,
  max_iter = 4,
  psi = "bisquare",
  c = 4.685
)

str(fit)
plot(dat$tps, fit$beta_hat, type = "l",
      xlab = "Functional domain",

```

```
ylab = expression(hat(beta)(s)))
```

rlfpca_estimate

Robust Longitudinal Functional Principal Component Analysis

Description

Estimates a robust longitudinal functional principal component decomposition for repeatedly observed functional predictors. The function robustly estimates the mean surface, subject-level and visit-level covariance components, eigenfunctions, eigenvalues, and longitudinal FPCA scores.

Usage

```
rlfpca_estimate(Y, subject, Time, L = 0.90,
               N.X = NA, N.U = NA,
               smooth = FALSE, bf = 8,
               df_time_mean = 5,
               mean_iter = 3,
               clean_cells = TRUE,
               clean_c = 4.5,
               score_iter = 8)
```

Arguments

Y	A numeric matrix containing the longitudinal functional predictor. Rows correspond to subject-visit observations and columns correspond to the functional grid points.
subject	A vector of subject identifiers. The length of subject must be equal to nrow(Y). Repeated observations from the same subject must have the same identifier.
Time	A numeric vector of visit times. The length of Time must be equal to nrow(Y). The time variable is internally standardized before estimating the longitudinal functional decomposition.
L	Target proportion of total variation explained by the retained subject-level and visit-level components when N.X or N.U is not supplied. The default is 0.90.
N.X	Number of subject-level longitudinal functional principal components to retain. If NA, the number is selected jointly with N.U using the pooled explained-variance rule determined by L. The default is NA.
N.U	Number of visit-level functional principal components to retain. If NA, the number is selected jointly with N.X using the pooled explained-variance rule determined by L. The default is NA.
smooth	Logical; if TRUE, the estimated covariance surfaces are smoothed using bivariate penalized spline smoothing. If FALSE, the raw robust covariance estimates are used, with the diagonal of the visit-level covariance estimated separately. The default is FALSE.
bf	Basis dimension used in the bivariate smoothing step for covariance surfaces. This argument is passed to the internal smoothing routines. The default is 8.
df_time_mean	Degrees of freedom used for the spline basis in the robust mean surface estimation step. The default is 5.

mean_iter	Maximum number of robust reweighting iterations used to estimate the mean surface. The default is 3.
clean_cells	Logical; if TRUE, unusually large cellwise residual values in the centered functional data matrix are winsorized before covariance estimation. The default is TRUE.
clean_c	Tuning constant used in the cellwise residual cleaning step. The default is 4.5.
score_iter	Maximum number of alternating blockwise score-update iterations used to estimate the subject-level and visit-level scores. The default is 8.

Details

The function implements the robust longitudinal functional principal component analysis step used in robust longitudinal functional principal component regression. Let $X_{ij}(s)$ denote the functional predictor for subject i at visit j , and let T_{ij} denote the corresponding visit time. The longitudinal functional predictor is represented as

$$X_{ij}(s) - \mu(s, T_{ij}) = B_{i0}(s) + t_{ij}B_{i1}(s) + U_{ij}(s) + \varepsilon_{ij}^X(s),$$

where $\mu(s, T_{ij})$ is the mean surface, $B_{i0}(s)$ is the subject-specific functional intercept process, $B_{i1}(s)$ is the subject-specific functional slope process, $U_{ij}(s)$ is the visit-specific functional deviation process, and $\varepsilon_{ij}^X(s)$ is a residual measurement-error process.

The function first estimates the mean surface robustly by iteratively weighted least squares. At each iteration, curvewise residual norms are computed and robust curve weights are updated. These weights reduce the influence of atypical functional observations when estimating the mean surface and the covariance components.

After centering the curves, the function estimates the subject-level and visit-level covariance components from the cross-product regression identity

$$E\{\tilde{X}_{ij}(s)\tilde{X}_{ik}(s') \mid T_{ij}, T_{ik}\} = G_0(s, s') + t_{ik}G_{10}(s, s') + t_{ij}G_{01}(s, s') + t_{ij}t_{ik}G_1(s, s') + I(j = k)G_U(s, s').$$

The kernels G_0 , G_1 , and G_{01} describe the subject-level functional intercept/slope covariance structure, whereas G_U describes the visit-level covariance structure. Robustness is introduced through pairwise weights formed from the curve weights.

The estimated subject-level covariance operator is constructed as the block operator

$$\hat{K}_X = \begin{pmatrix} \hat{K}_0 & \hat{K}_{01} \\ \hat{K}_{10} & \hat{K}_1 \end{pmatrix},$$

and the visit-level covariance operator is denoted by $\hat{K}_U$. Eigenanalysis of these two covariance operators yields subject-level eigenfunctions

$$\left\{ \left(\hat{\phi}_k^{(0)}, \hat{\phi}_k^{(1)} \right) : k = 1, \dots, K_X \right\}$$

and visit-level eigenfunctions

$$\left\{ \hat{\phi}_\ell^U : \ell = 1, \dots, K_U \right\}.$$

If $N.X$ and $N.U$ are not supplied, the function selects them sequentially by pooling the positive subject-level and visit-level eigenvalues and retaining components until the target explained variation L is reached.

Finally, the robust longitudinal scores are estimated from the truncated decomposition

$$\tilde{X}_{ij}(s) \approx \sum_{k=1}^{K_X} \xi_{ik} \{ \hat{\phi}_k^{(0)}(s) + t_{ij} \hat{\phi}_k^{(1)}(s) \} + \sum_{\ell=1}^{K_U} \zeta_{ij\ell} \hat{\phi}_\ell^U(s).$$

The score estimation is performed using alternating blockwise updates for the subject-level scores ξ_{ik} and visit-level scores $\zeta_{ij\ell}$. This avoids forming a single large global score system and exploits the longitudinal block structure of the data.

The returned object is designed to be passed directly to `fit_rlfpcr`, which uses the estimated scores and eigenfunctions in the robust scalar-on-function regression stage.

Value

A list with the following components:

<code>phi.0</code>	A matrix containing the estimated subject-level functional intercept eigenfunctions $\hat{\phi}_k^{(0)}$. Rows correspond to functional grid points and columns correspond to retained subject-level components.
<code>phi.1</code>	A matrix containing the estimated subject-level functional slope eigenfunctions $\hat{\phi}_k^{(1)}$. Rows correspond to functional grid points and columns correspond to retained subject-level components.
<code>phi.U</code>	A matrix containing the estimated visit-level eigenfunctions $\hat{\phi}_\ell^U$. Rows correspond to functional grid points and columns correspond to retained visit-level components.
<code>xi</code>	A matrix of estimated subject-level scores. Rows correspond to subjects and columns correspond to retained subject-level components.
<code>zeta</code>	A matrix of estimated visit-level scores. Rows correspond to subject-visit observations and columns correspond to retained visit-level components.
<code>N.X</code>	The number of retained subject-level longitudinal functional principal components.
<code>N.U</code>	The number of retained visit-level functional principal components.
<code>eta</code>	A matrix containing the estimated robust mean surface evaluated at the observed visit times and functional grid points. It has the same dimension as Y .
<code>Y.tilde</code>	The robustly centered functional data matrix, obtained after subtracting <code>eta</code> from Y . If <code>clean_cells = TRUE</code> , this component also reflects the cellwise residual cleaning step.
<code>sigma2</code>	Estimated residual measurement-error variance of the functional predictor.
<code>lambda</code>	Estimated eigenvalues of the retained subject-level covariance operator.
<code>nu</code>	Estimated eigenvalues of the retained visit-level covariance operator.
<code>curve_w</code>	Final robust curve weights obtained from the mean-surface estimation step.

Note

The theoretical results in the accompanying methodology are developed for fixed truncation dimensions K_X and K_U . In the implementation, N_X and N_U may also be selected adaptively using the pooled explained-variance rule controlled by L .

The function assumes that the functional predictors are densely observed on a common grid. Sparse or irregularly sampled functional data should be preprocessed before using this function.

See Also

[fit_rlfpcr](#), [simulate_data](#)

Examples

```
dat <- simulate_data(
  case = "1",
  nI = 100,
  nV = 4,
  tps = seq(0.5, 99.5, by = 1),
  sigma_x = 0.01,
  sigma_y = sqrt(2),
  sigma_b = sqrt(2),
  scalar_cov = 4,
  beta_scalar = c(0.6, 0.8, -0.5, 0.2),
  outlier = TRUE
)

X <- dat$X
y <- dat$y
subject <- dat$subject
times <- dat$times
tps <- dat$tps
scalar_cov <- dat$scalar_cov

lfpca <- rlfpc_estimate(
  Y = X,
  subject = subject,
  Time = times,
  L = 0.95,
  smooth = TRUE,
  bf = 8,
  df_time_mean = 5,
  mean_iter = 3,
  clean_cells = TRUE,
  clean_c = 4.5,
  score_iter = 8
)

str(lfpca)

matplot(dat$tps, lfpca$phi.0, type = "l",
  xlab = "Functional domain",
  ylab = "Estimated subject-level eigenfunctions")
```


simulate_data

*Simulate Longitudinal Functional Regression Data***Description**

Generates simulated data for robust longitudinal scalar-on-function regression. The function creates repeatedly observed functional predictors, repeated scalar responses, optional scalar explanatory variables, subject-level random effects, and optional structured outliers in the functional predictor and scalar response.

Usage

```
simulate_data(case = c("1", "2"),
              nI = 100,
              nV = 4,
              tps = seq(0.5, 99.5, by = 1),
              sigma_x = 0.01,
              sigma_y = 0.80,
              sigma_b = 0.60,
              lambda = c(2.0, 1.2, 0.7, 0.4),
              nu = c(1.2, 0.7, 0.4),
              scalar_cov = NULL,
              beta_scalar = c(0.6, 0.8, -0.5, 0.2),
              outlier = FALSE,
              prop_curve = 0.05,
              prop_y = 0.05,
              prop_subject = 0.02,
              amp_subject = 4,
              amp_visit = 4,
              amp_y = 5,
              seed = NULL)
```

Arguments

case	A character string specifying the simulation case. Available choices are "1" and "2". Case "1" uses a single global functional coefficient, whereas Case "2" uses separate subject-level and visit-level functional coefficient components.
nI	Number of subjects. The default is 100.
nV	Number of visits per subject. The current implementation uses a balanced longitudinal design with nV repeated observations for each subject. The default is 4.
tps	Numeric vector giving the grid points on which the functional predictor is observed. The default is seq(0.5, 99.5, by = 1).
sigma_x	Standard deviation of the measurement error added to the latent functional predictor. The default is 0.01.
sigma_y	Standard deviation of the scalar response error. The default is 0.80.
sigma_b	Standard deviation of the subject-specific random intercept in the scalar response model. The default is 0.60.

lambda	Numeric vector of eigenvalues controlling the variance of the subject-level longitudinal functional scores. The length of lambda determines the number of subject-level components generated. The default is $c(2.0, 1.2, 0.7, 0.4)$.
nu	Numeric vector of eigenvalues controlling the variance of the visit-level functional scores. The length of nu determines the number of visit-level components generated. The default is $c(1.2, 0.7, 0.4)$.
scalar_cov	Optional scalar explanatory variables. If NULL, no scalar explanatory variables are included. If a single integer between 1 and 4 is given, that number of scalar explanatory variables is generated internally. If a matrix or data frame is supplied, it is used as the scalar covariate matrix.
beta_scalar	Numeric vector of true scalar coefficients used when scalar explanatory variables are included. The default is $c(0.6, 0.8, -0.5, 0.2)$.
outlier	Logical; if TRUE, structured contamination is added to the data. The default is FALSE.
prop_curve	Proportion of subject-visit functional observations contaminated by visit-level functional outliers when outlier = TRUE. The default is 0.05.
prop_y	Proportion of scalar responses contaminated by response outliers when outlier = TRUE. The default is 0.05.
prop_subject	Proportion of subjects contaminated by subject-level functional outliers when outlier = TRUE. The default is 0.02.
amp_subject	Magnitude of the subject-level functional outlier perturbation. Larger values produce more severe subject-level functional contamination. The default is 4.
amp_visit	Magnitude of the visit-level functional outlier perturbation. Larger values produce more severe visit-level functional contamination. The default is 4.
amp_y	Magnitude of the scalar response outlier perturbation. Larger values produce more severe response contamination. The default is 5.
seed	Optional random seed for reproducibility. The default is NULL.

Details

The function generates data from a longitudinal scalar-on-function regression framework in which both the scalar response and the functional predictor are observed repeatedly over visits. For subject i and visit j , the general regression structure is

$$Y_{ij} = \alpha(T_{ij}) + q_{ij}^\top \delta + b_i + \int_{\mathcal{S}} \{X_{ij}(s) - \mu(s, T_{ij})\} \beta(s) ds + e_{ij},$$

where Y_{ij} is the scalar response, T_{ij} is the visit time, q_{ij} is an optional vector of scalar explanatory variables, δ is the corresponding scalar coefficient vector, b_i is a subject-specific random intercept, $X_{ij}(s)$ is the functional predictor, $\mu(s, T_{ij})$ is the longitudinal mean surface, and $\beta(s)$ is the functional coefficient.

The latent functional predictor is generated according to the longitudinal functional decomposition

$$W_{ij}(s) = \mu(s, T_{ij}) + \sum_{k=1}^{K_X} \xi_{ik} \{\phi_k^{(0)}(s) + t_{ij} \phi_k^{(1)}(s)\} + \sum_{\ell=1}^{K_U} \zeta_{ij\ell} \phi_\ell^U(s),$$

where ξ_{ik} are subject-level scores, $\zeta_{ij\ell}$ are visit-level scores, $\phi_k^{(0)}$ and $\phi_k^{(1)}$ represent subject-level intercept and slope basis functions, and ϕ_ℓ^U represents visit-level basis functions. The observed functional predictor is

$$X_{ij}(s_d) = W_{ij}(s_d) + \varepsilon_{ij}^X(s_d),$$

where $\varepsilon_{ij}^X(s_d)$ is independent Gaussian measurement error with standard deviation `sigma_x`.

In Case "1", the scalar response depends on the full latent functional signal through a single global coefficient function,

$$Y_{ij} = b_i + q_{ij}^\top \delta + 1.40 \int_S W_{ij}(s) \beta(s) ds + e_{ij}.$$

In Case "2", the response depends separately on the subject-level and visit-level functional components,

$$Y_{ij} = b_i + q_{ij}^\top \delta + 1.35 \int_S B_{ij}(s) \beta_B(s) ds + 1.10 \int_S U_{ij}(s) \beta_U(s) ds + e_{ij},$$

where $B_{ij}(s)$ denotes the subject-level longitudinal component and $U_{ij}(s)$ denotes the visit-level deviation component.

If `outlier = TRUE`, three types of structured contamination may be introduced. First, a subset of subjects receives a subject-level functional perturbation, affecting all visits for those subjects. Second, a subset of individual subject-visit curves receives a visit-level functional perturbation. Third, a subset of scalar responses receives heavy-tailed additive contamination. The clean versions of the functional predictor and scalar response are also returned, which allows direct comparison between clean and contaminated data.

The output of this function is designed to be used directly with `rlfpca_estimate` and `fit_rlfpcr`.

Value

A list with the following components:

<code>case</code>	The simulation case used, either "1" or "2".
<code>nI</code>	Number of subjects.
<code>n</code>	Total number of subject-visit observations, equal to <code>nI * nV</code> .
<code>Jvec</code>	A vector containing the number of visits for each subject.
<code>subject</code>	Vector of subject identifiers of length <code>n</code> .
<code>times</code>	Vector of standardized visit times of length <code>n</code> .
<code>tps</code>	Functional grid points on which the curves are observed.
<code>X</code>	Observed functional predictor matrix. Rows correspond to subject-visit observations and columns correspond to functional grid points. If <code>outlier = TRUE</code> , this matrix contains the contaminated functional observations.
<code>y</code>	Observed scalar response vector. If <code>outlier = TRUE</code> , this vector contains the contaminated scalar responses.
<code>X_clean</code>	Clean functional predictor matrix before structured outliers are added.
<code>y_clean</code>	Clean scalar response vector before response outliers are added.
<code>scalar_cov</code>	Matrix of scalar explanatory variables, or <code>NULL</code> if no scalar explanatory variables are used.
<code>beta_scalar_true</code>	True scalar coefficient vector used to generate the scalar covariate effect.
<code>scalar_names</code>	Names of the scalar explanatory variables.

<code>mu_true</code>	True conditional mean of the scalar response before adding response noise.
<code>W_latent</code>	Latent functional signal before adding functional measurement error.
<code>B_comp</code>	Subject-level longitudinal functional component.
<code>U_comp</code>	Visit-level functional deviation component.
<code>eta_mat</code>	Mean surface component evaluated at the observed visit times and functional grid points.
<code>truth</code>	A list containing the true coefficient functions. In Case "1", <code>truth\$beta_true</code> contains the global coefficient function, while <code>truth\$beta_B_true</code> and <code>truth\$beta_U_true</code> are NULL. In Case "2", <code>truth\$beta_B_true</code> and <code>truth\$beta_U_true</code> contain the subject-level and visit-level coefficient functions, while <code>truth\$beta_true</code> is NULL.
<code>outlier</code>	Logical indicator showing whether structured contamination was added.
<code>idx_subject_out</code>	Indices of subject-visit observations belonging to contaminated subjects. This is an empty integer vector when <code>outlier = FALSE</code> .
<code>idx_visit_out</code>	Indices of subject-visit observations contaminated by visit-level functional outliers. This is an empty integer vector when <code>outlier = FALSE</code> .
<code>idx_y_out</code>	Indices of scalar responses contaminated by response outliers. This is an empty integer vector when <code>outlier = FALSE</code> .

Note

The current implementation generates balanced longitudinal data with the same number of visits for each subject. The functional predictor is observed densely on a common grid `tps`. The function is intended primarily for simulation studies, reproducible examples, and validation of the RLFPCR estimation pipeline.

See Also

[rlfpca_estimate](#), [fit_rlfpcr](#)

Examples

```
dat <- simulate_data(
  case = "1",
  nI = 100,
  nV = 4,
  tps = seq(0.5, 99.5, by = 1),
  sigma_x = 0.01,
  sigma_y = sqrt(2),
  sigma_b = sqrt(2),
  scalar_cov = 4,
  beta_scalar = c(0.6, 0.8, -0.5, 0.2),
  outlier = TRUE
)

str(dat)

matplot(dat$tps, t(dat$X[1:5, ]), type = "l",
  xlab = "Functional domain",
  ylab = "Functional predictor")
```

```
plot(dat$tps, dat$truth$beta_true, type = "l",  
     xlab = "Functional domain",  
     ylab = "True coefficient function")
```

```
dat_out <- simulate_data(  
  case = "2",  
  nI = 100,  
  nV = 4,  
  tps = seq(0.5, 99.5, by = 1),  
  sigma_x = 0.01,  
  sigma_y = 0.80,  
  sigma_b = 0.60,  
  lambda = c(2.0, 1.2, 0.7, 0.4),  
  nu = c(1.2, 0.7, 0.4),  
  scalar_cov = 4,  
  beta_scalar = c(0.6, 0.8, -0.5, 0.2),  
  outlier = FALSE  
)
```

```
length(dat_out$idx_subject_out)  
length(dat_out$idx_visit_out)  
length(dat_out$idx_y_out)
```

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